Synoptic LE: Tree Fluxes 2022–2023

Roberta Peixoto

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1 Read in and collate all the files in the Processed Data folder

2 Remove bad fluxes and calculate how many there were

```
##Remove any samples with "remove" in the qaqc_flag columns
# stem_flux_2022_2023 <- stem_flux_2022_2023 %>%
#   filter(!grepl("remove", ch4_qaqc_flag, ignore.case = TRUE)) %>%
#   filter(!grepl("remove", co2_qaqc_flag, ignore.case = TRUE))

##Removed fluxes in the CH4 and CO2 flux processing Rmd code, removed both CH4 and CO2 obs w
```

```r
r2_summary <- stem_flux_2022_2023 %>%
                      summarise(total_obs = n(),
                                n_neg_CO2 = sum(CO2_neg_flux , na.rm = TRUE),
                                pct_neg_CO2 = 100 * n_neg_CO2 / total_obs,

                                n_CO2_flux_rejected = sum(CO2_flux_rejected),
                                pct_CO2_flux_rejected = 100 * n_CO2_flux_rejected / total_obs

                                n_CH4_rejected_model = sum(CH4_rejected_model),
                                pct_CH4_rejected_model = 100 * n_CH4_rejected_model / total_

#table
r2_summary %>%
  tidyr::pivot_longer(everything(), names_to = "Metric", values_to = "Value") %>%
  mutate(
    Metric = recode(Metric,
      total_obs              = "Total observations",
      n_neg_CO2              = "Negative CO2 flux (n)",
      pct_neg_CO2            = "Negative CO2 flux (%)",
      n_CO2_flux_rejected    = "Rejected CO2 flux (n)",
      pct_CO2_flux_rejected  = "Rejected CO2 flux (%)",
      n_CH4_rejected_model   = "Rejected CH4 - poor model fit (n)",
      pct_CH4_rejected_model = "Rejected CH4 - poor model fit (%)"
    ),
    Value = ifelse(grepl("%", Metric),
                   round(Value, 1),
                   as.integer(Value))
  ) %>%
  knitr::kable(
    caption  = "QA/QC CO2 and CH4 Flux Summary",
    align    = c("l", "r"),
    booktabs = TRUE,
    format   = "latex"
  ) %>%
  kableExtra::kable_styling(
    latex_options = c("striped", "hold_position"),
    full_width    = FALSE
  ) %>%
  kableExtra::pack_rows("CO2", 2, 5) %>%
  kableExtra::pack_rows("CH4", 6, 7)
```

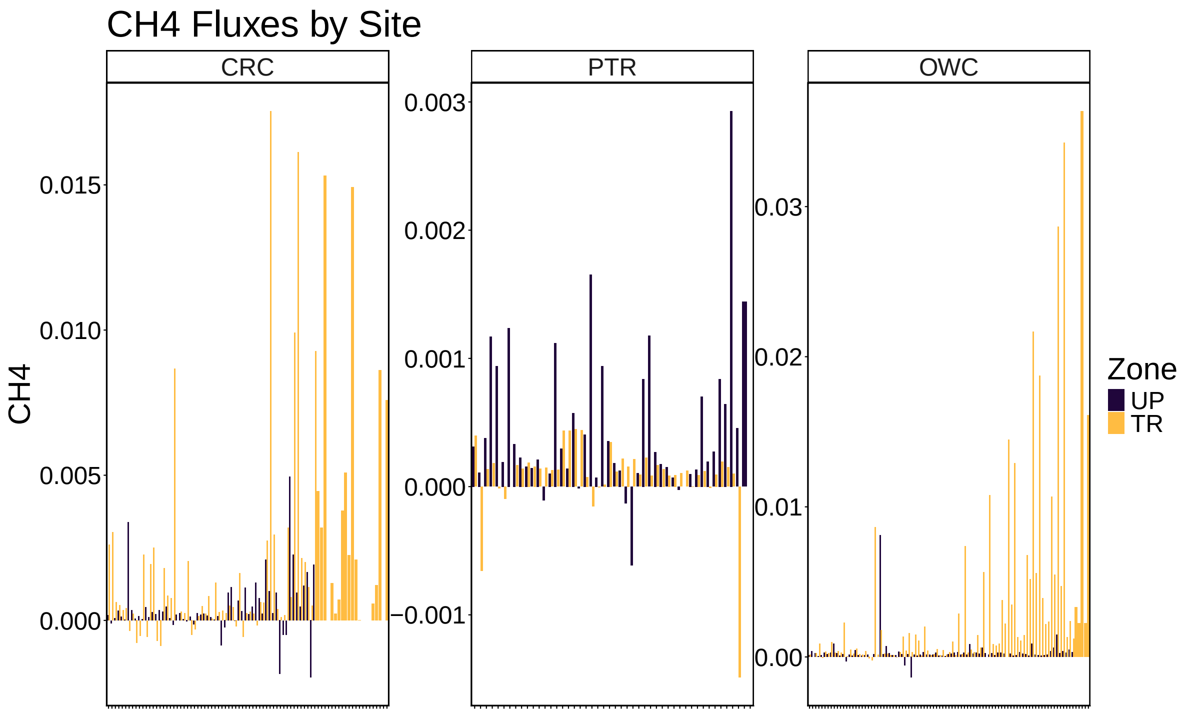
Table 1: QA/QC CO2 and CH4 Flux Summary

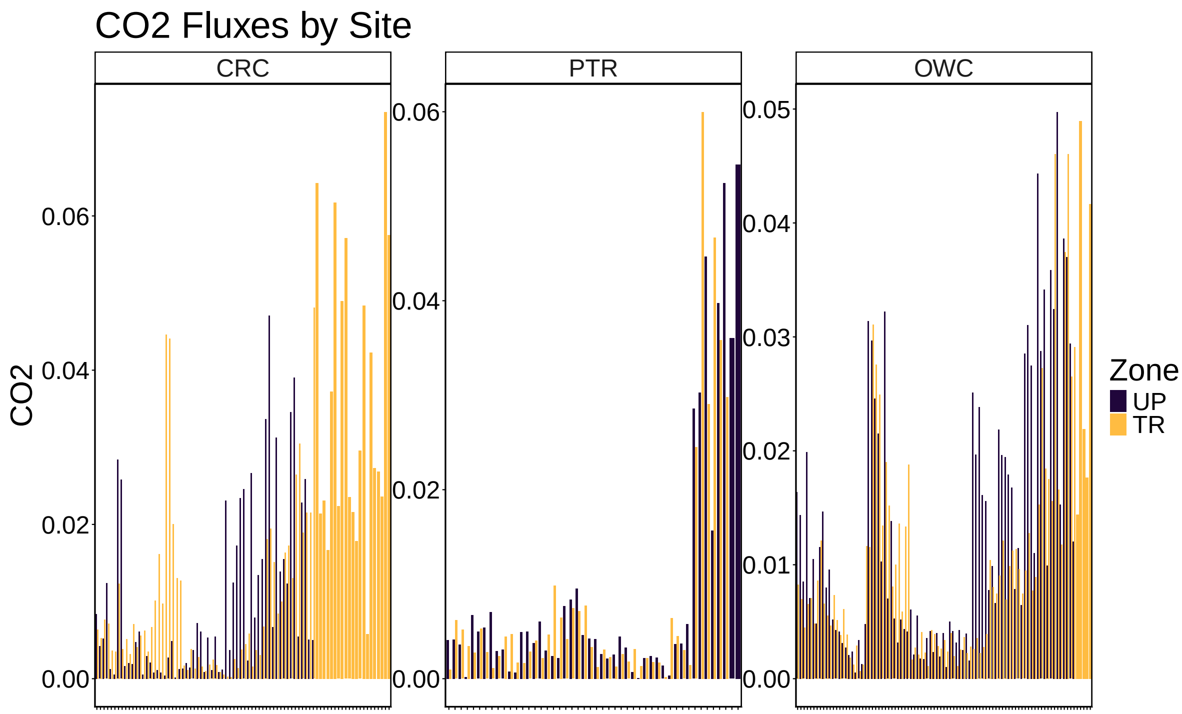
Metric	Value
Total observations	424.0
CO2	
Negative CO2 flux (n)	10.0
Negative CO2 flux (%)	2.4
Rejected CO2 flux (n)	12.0
Rejected CO2 flux (%)	2.8
CH4	
Rejected CH4 — poor model fit (n)	NA
Rejected CH4 — poor model fit (%)	NA

```
#removing negative CO2 fluxes from the final dataset but leaving flags for users to decide
stem_flux_2022_2023_filtered <-stem_flux_2022_2023%>%
                      filter(co2_lin_flux.estimate>0)

##note that this list has some not synoptic monitoring samples zone-labeled as "Luci" that w

##note that this list has some diurnal variation samples for PTR synoptic veg campaign in 20
```

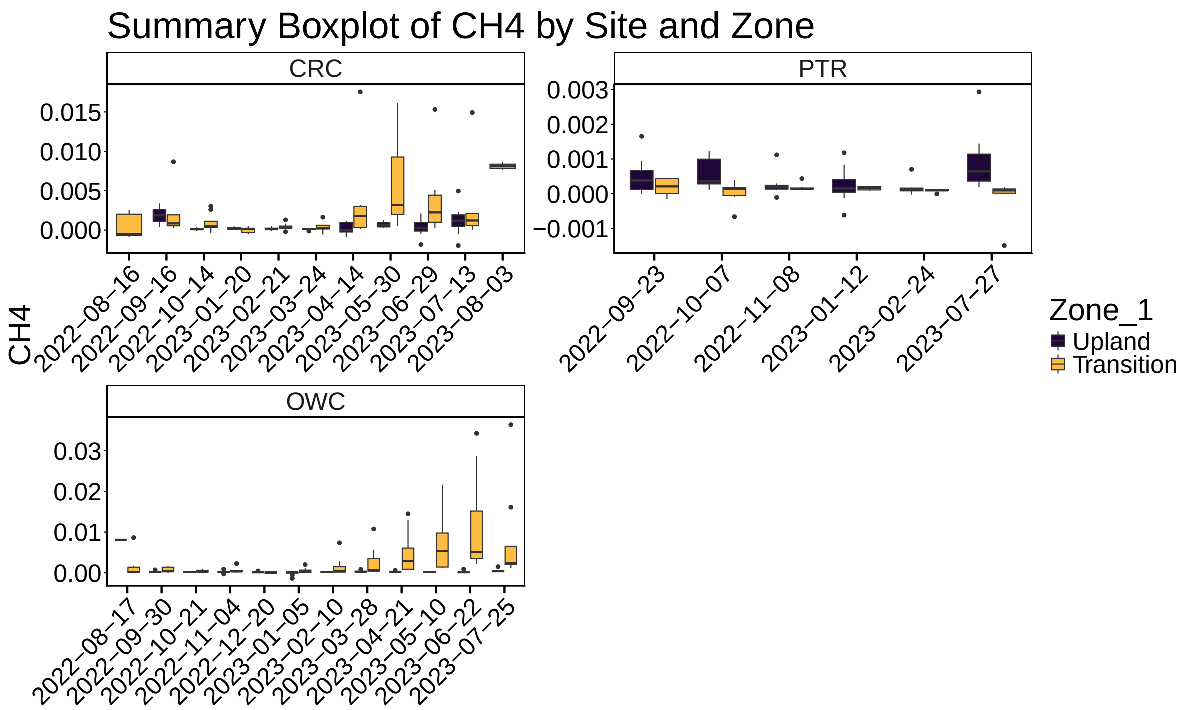
3 Write out files

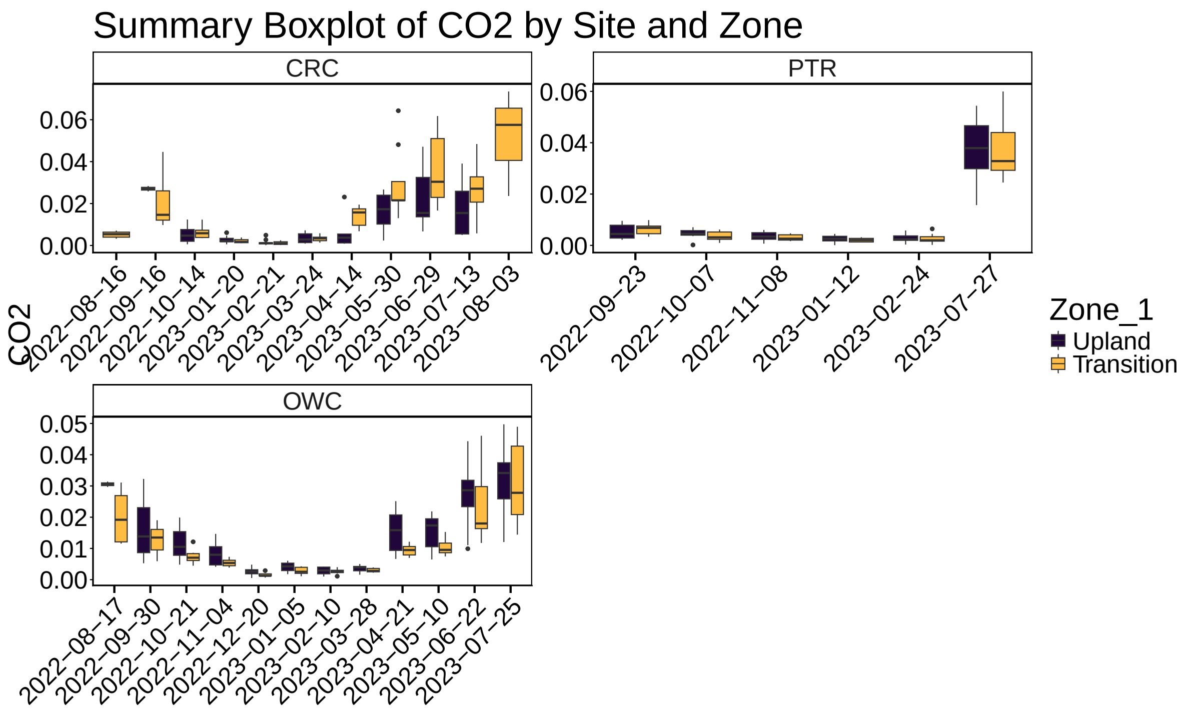
4 All Data by Year, Site and Zone



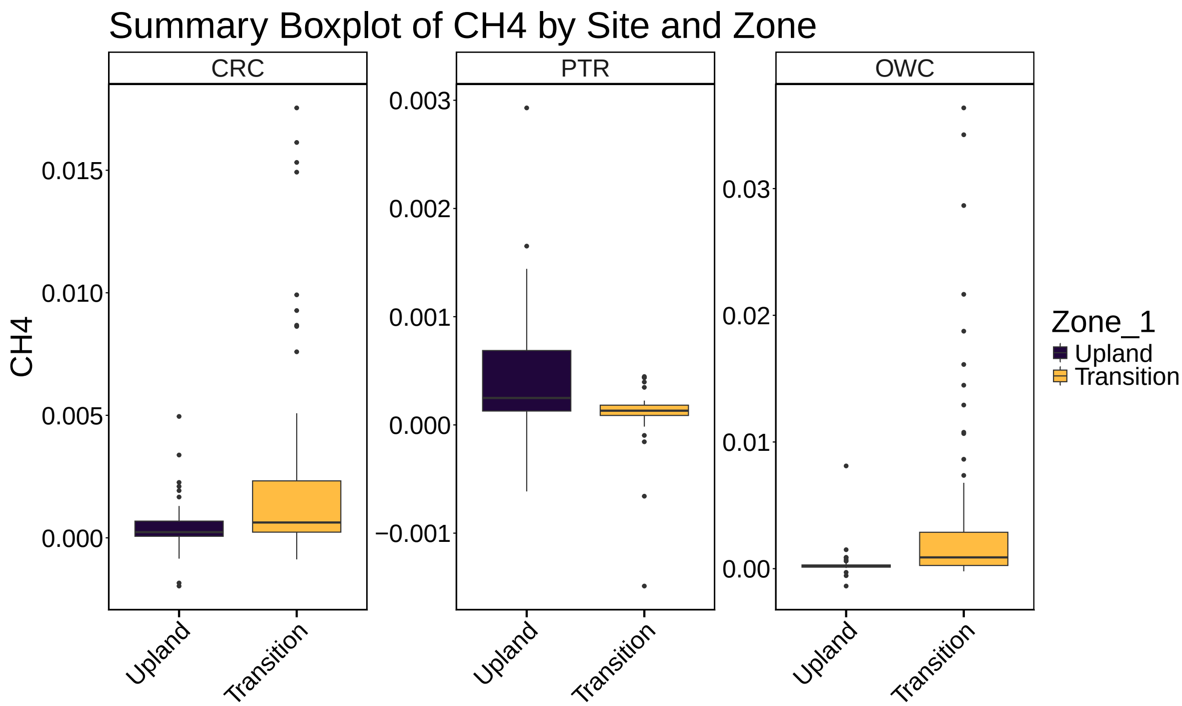

CO2 Fluxes by Site
CRC
PTR
OWC
CO2
0.00
0.02
0.04
0.06
0.00
0.02
0.04
0.06
0.00
0.01
0.02
0.03
0.04
0.05
Zone
UP
TR

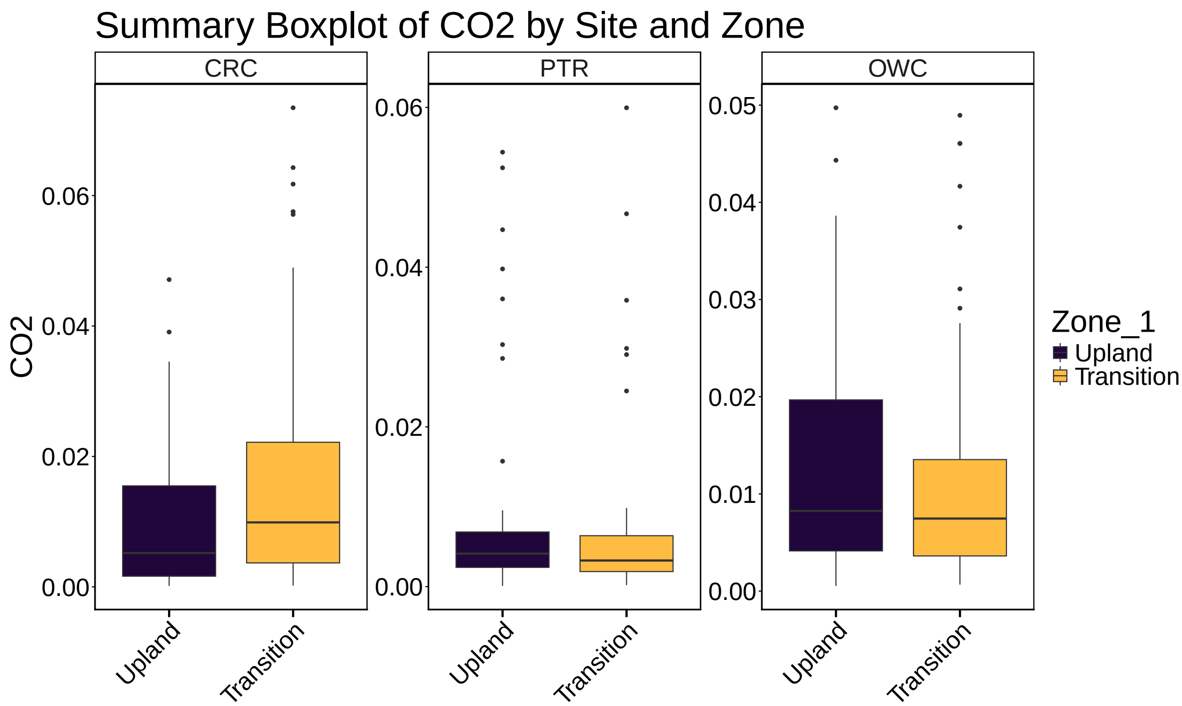
5 Boxplot summary by Date




Summary Boxplot of CO2 by Site and Zone
CRC
PTR
OWC
CO2
0.00
0.01
0.02
0.03
0.04
0.05
0.06
2022−08−16
2022−09−16
2022−10−14
2023−01−20
2023−02−21
2023−03−24
2023−04−14
2023−05−30
2023−06−29
2023−07−13
2023−08−03
2022−09−23
2022−10−07
2022−11−08
2023−01−12
2023−02−24
2023−07−27
2022−08−17
2022−09−30
2022−10−21
2022−11−04
2022−12−20
2023−01−05
2023−02−10
2023−03−28
2023−04−21
2023−05−10
2023−06−22
2023−07−25
Zone_1
Upland
Transition

6 Boxplot summary by Site





7 Calculate summary metrics

8 Summary Tables

Table 2: Summary Statistics by Site and Zone

Site	Zone	Min CH4	Max CH4	Mean CH4	Min CO2	Max CO2	Mean CO2
CRC	Upland	-0.0019672	0.0049544	0.0004504	0.0001479	0.0471143	0.0102611
CRC	Transition	-0.0008791	0.0175446	0.0022702	0.0001944	0.0734572	0.0164906
PTR	Upland	-0.0006145	0.0029295	0.0004562	0.0000843	0.0544040	0.0093565
PTR	Transition	-0.0014889	0.0004471	0.0000943	0.0001934	0.0599590	0.0079690
OWC	Upland	-0.0013717	0.0081095	0.0003176	0.0005345	0.0497277	0.0131000
OWC	Transition	-0.0002087	0.0363698	0.0035907	0.0006682	0.0489540	0.0109439